

Algorithmic Theory of Qubit Routing in the Linear Nearest Neighbor Architectures

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The qubit routing problem, also known as the swap minimization problem, is a (classical) combinatorial optimization problem that arises in the design of compilers of quantum programs. We study the qubit routing problem from the viewpoint of theoretical computer science, while most of the existing studies investigated the practical aspects. We concentrate on the linear nearest neighbor (LNN) architectures of quantum computers, in which the graph topology is a path. Our results are three-fold. (1) We prove that the qubit routing problem is NP-hard. (2) We show that the qubit routing problem on a path can be solved in a fixed-parameter time where the number of two-qubit gates is a parameter. (3) We show that the qubit routing problem on a path can be solved in polynomial time if each qubit is involved in at most one two-qubit gate.

CCS Concepts: • **Theory of computation → Fixed parameter tractability**; • **Computer systems organization → Quantum computing;**

Additional Key Words and Phrases: Qubit routing, Qubit allocation, Fixed-parameter tractability

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1 Introduction

The qubit routing problem captures a (classical) combinatorial problem in designing compilers of quantum programs. We rely on the formulation introduced by Siraichi et al. [11]. In their setting,

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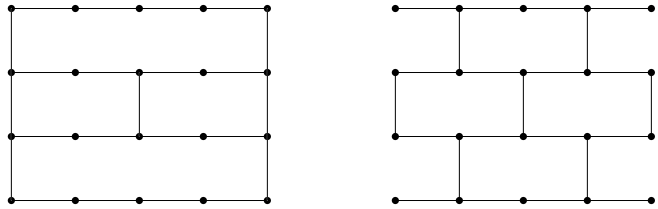


Fig. 1. The graph topology of IBM quantum devices called "Johannesburg" (left) and "Almaden" (right). Each vertex represents a physical qubit, and we may perform a two-input gate operation only for a pair of adjacent qubits. In our problem formulation, the graph topology is taken into account as the graph G. Source: https://www.ibm.com/blogs/research/2019/09/quantum-computation-center/.

a quantum program is designed as a quantum circuit. In a quantum circuit, there are wires and quantum gates such as Hadamard gates and controlled NOT gates. Each wire corresponds to one quantum bit (or a qubit for short), and gates operate on one or more qubits at a time. A quantum circuit is designed at the logic level and needs to be implemented at the physical level. This requires a mapping of logical qubits to physical qubits in such a way that all the gate operations can be performed. However, due to physical restrictions, some sets of qubits could be mapped to places where an operation on those qubits cannot be physically performed. This problem is essential for some of the superconducting quantum computers such as IBM Quantum systems. Figure 1 shows the graph topology of such computers.

To overcome this issue, we insert so-called swap gates into the quantum circuit. A swap gate can swap two qubits on different wires and can be implemented by a sequence of three controlled NOT gates. With swap operations, we will be able to perform all the necessary gate operations that were designed in the original circuit. The process is divided into two phases. In the first phase, we find an initial mapping of logical qubits to physical qubits. In the second phase, we determine how swap operations are performed. Since a swap operation incurs a cost, we want to minimize the number of swap operations. The quantum routing problem focuses on the second phase of the process.

Several methods have been proposed to solve the quantum routing problem. The existing research has mainly focused on the practical aspects of the problem. However, the theoretical investigation has been scarce. Some authors have claimed that the problem is NP-hard, but no formal proof was given to date. Until this article, it was not known whether the problem can be solved in polynomial time for the simplest case where the physical placement of qubits forms a path and the program has no two gate operations that can be performed at the same time. A path corresponds to the case of the linear nearest neighbor architectures that have been extensively studied in the literature of quantum computing (e.g., Reference [10]).

This article focuses on the theoretical aspects of the quantum routing problem and gives a better understanding of the problem from the viewpoint of theoretical computer science. We are mainly concerned with the case where the physical placement of qubits forms a path. Under this restriction, we obtain the following results.

(1) We prove that the qubit routing problem is NP-hard even when the program has no two gate operations that can be performed at the same time (Theorem 1).
(2) We give a fixed-parameter algorithm for the qubit routing problem when the number of gates in a given quantum circuit is a parameter (Theorem 4).
(3) We give a polynomial-time algorithm for the qubit routing problem when each qubit is involved in at most one two-qubit operation (Theorem 5).

Table 1. Summary of Our Results

Graphs	setting on 2-gates operations	our results
path	total order	FPT w.r.t. # of 2-gates operations (Theorem 4)
	(partial order)	NP-hard (Theorem 1)
	disjoint pairs	polynomial-time solvable (Theorem 5)
star	anti-chain	NP-hard (Theorem 6)

As a side result, we also prove that the problem is NP-hard when the physical placement of qubits forms a star and any set of gate operations can be performed at the same time (Theorem 6). See Table 1.

This article concentrates on theory and complements contributions of papers in the existing literature that mostly concern practical aspects of the qubit routing problem; see Related Work part at the end of this section. We here emphasize that theoretical studies and practical studies play complementary roles in the research on algorithms. Practical studies are usually conducted with implementation and experiments and have more direct implications for real-world problem solving. However, evaluations are limited, since experimental setups cannot be exhaustive. Theoretical studies are usually conducted with mathematical proofs and provide more solid foundations for all the cases to which theorems are applied. However, mathematical theorems are usually accompanied by several assumptions that may limit their applicability. In the case of the qubit routing problem, theoretical studies have been lacking. Thus, this article initiates theoretical studies of this important problem.

Problem Formulation. We formulate the problem Qubit Routing in a purely combinatorial way as follows. As input, we are given an undirected graph $G = (V, E)$, a partially ordered set (a poset for short) $P = (S, \preceq)$, a set T of tokens, a map $\varphi\colon S \to \binom{T}{2}$, where $\binom{T}{2}$ denotes the set of unordered pairs of T, and an initial token placement $f_0\colon V \to T$, which is defined as a bijection. The undirected graph G corresponds to the physical architecture of a quantum computer in which each vertex corresponds to a physical qubit and an edge corresponds to a pair of qubits on which a gate operation can be performed. The poset P corresponds to a quantum circuit that we want to run on the quantum computer. Each token in T corresponds to a logical qubit. The token placement f_0 corresponds to an initial mapping of the logical qubits in T to the physical qubits in V (e.g., as a result of the qubit allocation; see "Related Work" below). For the notational convenience, we regard f_0 as a mapping from V to T; this is not mathematically relevant, since f_0 is bijective and to construct a mapping from T to V, one uses the inverse f_0^{-1}. The bijectivity of a token placement is irrelevant, since if there are more logic qubits than physical qubits, then we may introduce dummy logical qubits so that their numbers can be equal. Each element in S corresponds to the pair of logical qubits on which a gate operation is performed. The correspondence is given by φ. We note that φ does not have to be injective. The poset P determines the order along which the gate operations determined by φ are performed. The order is partial, since some pairs of operations may be performed independently: In that case, the corresponding elements of S are incomparable in P. An example is given in Figure 2.

A *swap* $f_i \rightsquigarrow f_{i+1}$ is defined as an operation that transforms one token placement (i.e., a bijection) $f_i\colon V \to T$ to another token placement $f_{i+1}\colon V \to T$ such that there exists an edge $\{u, v\} \in E$ such that $f_{i+1}(u) = f_i(v)$, $f_{i+1}(v) = f_i(u)$ and $f_{i+1}(x) = f_i(x)$ for all $x \in V \setminus \{u, v\}$. This corresponds to a swap operation of two logical qubits $f_i(u)$ and $f_i(v)$ assigned to two different physical qubits u and v.

As an output of the qubit routing problem, we want a swap sequence $f_0 \rightsquigarrow f_1 \rightsquigarrow \cdots \rightsquigarrow f_\ell$ that satisfies the following condition. There exists a map $i\colon S \to \{0, 1, 2, \ldots, \ell\}$ such that $f_{i(s)}^{-1}(\varphi(s)) \in E$

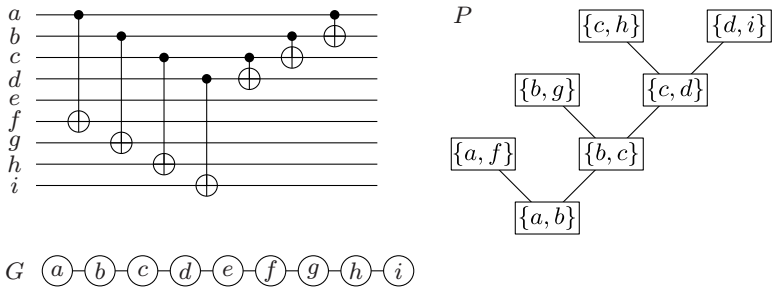


Fig. 2. The upper-left figure shows an instance of a quantum circuit, which is a part of the circuits given in Reference [2]. The right figure shows the Hasse diagram of the corresponding poset P in our problem formulation. The lower-left figure is an initial token placement on a graph (a path) G corresponding to the physical architecture of a quantum computer.

for every $s \in S$, and if $s \preceq s'$, then $i(s) \leq i(s')$. A swap sequence with this condition is said to be *feasible*. The objective is to minimize the length ℓ of a swap sequence. The condition states that all the operations in the quantum circuit that corresponds to the poset P are performed as they follow the order P.

In summary, our problem is described as follows.

Qubit Routing

Input. A graph $G = (V, E)$, a poset $P = (S, \preceq)$, a set T of tokens, a map $\varphi\colon S \to \binom{T}{2}$, and an initial token placement $f_0\colon V \to T$.

Question. Find a feasible swap sequence of minimum length.

Note that the minimum length of a feasible swap sequence is at most $|V||S|$ (if G is connected), which implies that determining the existence of a feasible swap sequence of length bounded by a given number is in NP.

In Figure 2, since each pair $\{a, b\}, \{b, c\}, \{c, d\}$ in P is adjacent in the initial token placement of G, the corresponding gates can be performed without swapping. However, to perform the gate determined by $\{d, i\}$, we need to move the tokens d and/or i by swap operations so that they are adjacent in G.

Related Work. The qubit allocation problem was introduced by Siraichi et al. [11]. Following Nannicini et al. [9], we divide the task in the qubit allocation problem into two phases. The first phase involves qubit assignment that gives an initial placement of the logical qubits on the physical qubits, and the second phase involves qubit routing that inserts swap operations at appropriate positions so that all the gate operations in a given circuit can be performed. The problem formulation in this article solves the second-phase problem. The qubit routing problem is often called the swap minimization, too.

For qubit routing, several heuristic algorithms have been given [7, 11, 12], and integer-programming formulations have been given [9, 13] that attempt to solve problem instances to optimality. Siraichi et al. [11, 12] have claimed that the qubit routing problem is NP-hard, since it is more general than the so-called token swapping problem [14] that is known to be NP-hard [3, 6, 8] even for trees [1]. However, their argument was informal and no formal proof was found.

A similar problem was studied by Botea, Kishimoto, and Marinescu [4]. In their problem, a mapping of logical qubits to physical qubits is injective but not bijective, and a logical qubit can

only be moved to a physical qubit that is not occupied by another logical qubit. They proved that the makespan minimization in their setting is NP-hard.

2 Hardness: Paths and Chains

In this section, we show that the problem is NP-hard even when G is a path and P is a chain (i.e., a totally ordered set).

Theorem 1. *Qubit Routing is NP-hard even when G is a path and P is a chain.*

To prove Theorem 1, we first introduce notation. For a swap sequence $f_0 \rightsquigarrow f_1 \rightsquigarrow \cdots \rightsquigarrow f_\ell$, which is denoted by $\mathbf{f}$, we say that f_0 and f_ℓ are the *initial* and *target* token placements of $\mathbf{f}$, respectively. We also say that $\mathbf{f}$ is from f_0 to f_ℓ. The length ℓ of $\mathbf{f}$ is denoted by $|\mathbf{f}|$. For two swap sequences $\mathbf{f}_1$ and $\mathbf{f}_2$, if the target token placement of $\mathbf{f}_1$ is equal to the initial token placement of $\mathbf{f}_2$, then its concatenation is denoted by $\mathbf{f}_1 \circ \mathbf{f}_2$. Note that in the concatenation of swap sequences, the target token placement of $\mathbf{f}_1$ and the initial token placement of $\mathbf{f}_2$ are identified, and thus $|\mathbf{f}_1 \circ \mathbf{f}_2| = |\mathbf{f}_1| + |\mathbf{f}_2|$. When the initial and the target token placements of $\mathbf{f}$ coincide, for a positive integer h, we denote $\mathbf{f}^h = \mathbf{f} \circ \mathbf{f} \circ \cdots \circ \mathbf{f}$, where $\mathbf{f}$ appears h times.

Throughout this section, we only consider the case where P is a chain. For a chain $P = (S, \preceq)$, let $s_1, s_2, \ldots, s_{|S|}$ be distinct elements in S such that $s_1 \prec s_2 \prec \cdots \prec s_{|S|}$. Then, the information of P and φ can be represented as a sequence $Q = (q_1, q_2, \ldots, q_{|S|})$, where $q_i := \varphi(s_i) \in \binom{T}{2}$ for each i. We say that a swap sequence $f_0 \rightsquigarrow f_1 \rightsquigarrow \cdots \rightsquigarrow f_\ell$ *realizes* Q if there exist $0 \le i_1 \le i_2 \le \cdots \le i_{|S|} \le \ell$ such that $f_{i_j}^{-1}(q_j) \in E$ for $j = 1, 2, \ldots, |S|$. In particular, if the swap sequence consisting of a single token placement f realizes Q, then we say that f realizes Q. With this terminology, when P is a chain, Qubit Routing is to find a shortest swap sequence that realizes a given sequence of token pairs. For two sequences Q_1 and Q_2 of token pairs, its concatenation is denoted by $Q_1 \circ Q_2$. For a sequence Q of token pairs and a positive integer h, we denote $Q^h = Q \circ Q \circ \cdots \circ Q$, where Q appears h times.

To show the NP-hardness of Qubit Routing, we reduce Optimal Linear Arrangement, which is known to be NP-hard [5].

> Optimal Linear Arrangement
>
> **Input.** A graph $H = (V(H), E(H))$ and a positive integer k.
> **Question.** Is there a bijection $g\colon V(H) \to \{1, 2, \ldots, |V(H)|\}$ that satisfies $\sum_{\{u,v\} \in E(H)} |g(u) - g(v)| \le k$?

Suppose that we are given an instance of Optimal Linear Arrangement that consists of a graph $H = (V(H), E(H))$ and a positive integer k. Denote $V(H) = \{v_1, v_2, \ldots, v_n\}$ and $E(H) = \{e_1, e_2, \ldots, e_m\}$, where $n = |V(H)|$ and $m = |E(H)|$. We may assume that $k < nm$, since, otherwise, any bijection g is a solution to Optimal Linear Arrangement. Let $\alpha = 2nm + 1$, which is an odd integer, and let β and γ be sufficiently large integers (e.g., $\beta = n^2\alpha$ and $\gamma = 4k\alpha$).

We now construct an instance of Qubit Routing as follows. Define a set of tokens as $T = \{t_{v,i} \mid v \in V(H),\ i \in \{1, 2, \ldots, \alpha\}\}$. Let $G = (V, E)$ be a path with $n\alpha$ vertices. We define

$$
\begin{aligned}
Q_v &:= (\{t_{v,1}, t_{v,2}\}, \{t_{v,2}, t_{v,3}\}, \ldots, \{t_{v,\alpha-1}, t_{v,\alpha}\}) && (v \in V(H)),\\
Q &:= Q_{v_1} \circ Q_{v_2} \circ \cdots \circ Q_{v_n},\\
\psi(e) &:= (\{t_{u,nm+1}, t_{v,nm+1}\}) && (e = \{u, v\} \in E(H)),\\
R &:= Q^\gamma \circ \psi(e_1) \circ Q^\gamma \circ \psi(e_2) \circ \cdots \circ \psi(e_m) \circ Q^\gamma,
\end{aligned}
$$

and R^β is the sequence of token pairs that has to be realized. The initial token placement f_0 is defined arbitrarily. This gives an instance (G, T, R^β, f_0) of Qubit Routing.

To show the validity of the reduction, it suffices to show the following.

Proposition 1. *The above instance (G, T, R^β, f_0) of Qubit Routing has a solution of length less than $2\beta(k\alpha + \alpha - m)$ if and only if the original instance of Optimal Linear Arrangement has a solution of objective value at most k.*

2.1 Proof of Proposition 1

To simpify the notation, we regard the vertex set V of G as $\{1, 2, \ldots, n\alpha\}$ so that $\{i, i+1\} \in E$ for $i \in \{1, 2, \ldots, n\alpha - 1\}$. We say that a token placement $f\colon V \to T$ is *block-aligned* if $|f^{-1}(t_{v,i}) - f^{-1}(t_{v,i+1})| = 1$ for any $v \in V$ and for any $i \in \{1, 2, \ldots, \alpha - 1\}$. In other words, f is block-aligned if and only if, for any $v \in V(H)$, the vertices $f^{-1}(t_{v,1}), f^{-1}(t_{v,2}), \ldots, f^{-1}(t_{v,\alpha})$ appear consecutively in this order (or in the reverse order) on the path G. Observe that f is block-aligned if and only if it realizes Q. In what follows, we show the sufficiency and the necessity in Proposition 1 separately.

Sufficiency ("if" part). Suppose that there exists a bijection $g\colon V(H) \to \{1, 2, \ldots, |V(H)|\}$ such that $\sum_{\{u,v\}\in E(H)} |g(u) - g(v)| \le k$. Define a token placement $f^*\colon V \to T$ as $f^*((g(v)-1)\alpha + i) = t_{v,i}$ for $v \in V(H)$ and $i \in \{1, 2, \ldots, \alpha\}$. Then, f^* is block-aligned, and hence f^* realizes Q. This implies that f^* realizes Q^γ, too.

We use the following two lemmas.

Lemma 1. *There exists a swap sequence $\mathbf{f}_0$ from f_0 to f^* whose length is less than $(n\alpha)^2$.*

Proof. Since $|V| = |T| = n\alpha$, by applying swaps at most $n\alpha - 1$ times to f_0, we obtain a token placement f_1 such that an end vertex v of the path G satisfies $f_1(v) = f^*(v)$. By applying the same operation for each vertex v in G from one end to the other, we obtain f^*. The total number of token swaps is at most $(n\alpha - 1)|V| < (n\alpha)^2$. □

Lemma 2. *For any $e = \{u, v\} \in E(H)$, there exists a swap sequence $\mathbf{f}_e$ from f^* to f^* such that $\mathbf{f}_e$ realizes $\psi(e) = (\{t_{u,nm+1}, t_{v,nm+1}\})$ and $|\mathbf{f}_e| = 2|g(u) - g(v)|\alpha - 2$.*

Proof. By applying swaps $|(f^*)^{-1}(t_{u,nm+1}) - (f^*)^{-1}(t_{v,nm+1})| - 1 = |g(u) - g(v)|\alpha - 1$ times to f^*, we can obtain a token placement f_e such that $(f_e)^{-1}(t_{u,nm+1})$ and $(f_e)^{-1}(t_{v,nm+1})$ are adjacent, i.e., f_e realizes $\psi(e)$. Conversely, f_e can be transformed to f^* by applying $|g(u) - g(v)|\alpha - 1$ swaps. Therefore, there exists a swap sequence of length $2|g(u) - g(v)|\alpha - 2$ that contains f^*, f_e, f^* in this order, which completes the proof. □

We now show that the following swap sequence satisfies the conditions.

$$\mathbf{f} := \mathbf{f}_0 \circ \left(\mathbf{f}_{e_1} \circ \mathbf{f}_{e_2} \circ \cdots \circ \mathbf{f}_{e_m}\right)^\beta,$$

where $\mathbf{f}_0$ and $\mathbf{f}_{e_i}$ are as in Lemmas 1 and 2, respectively. Since $\mathbf{f}_{e_1} \circ \mathbf{f}_{e_2} \circ \cdots \circ \mathbf{f}_{e_m}$ realizes R, we see that $\mathbf{f}$ realizes R^β. Furthermore, we obtain

$$\begin{aligned}
|\mathbf{f}| &= |\mathbf{f}_0| + \beta \sum_{i=1}^{m} |\mathbf{f}_{e_i}| \\
&< (n\alpha)^2 + \beta \sum_{\{u,v\}\in E(H)} (2|g(u) - g(v)|\alpha - 2) \\
&\le (n\alpha)^2 + 2\beta(k\alpha - m) \\
&< 2\beta(\alpha + k\alpha - m).
\end{aligned}$$

This shows that $\mathbf{f}$ is a desired swap sequence.

Necessity ("only if" part). To show the necessity, we first show a few properties of block-aligned token placements.

LEMMA 3. *If a token placement $f\colon V \to T$ is block-aligned, then there exists a bijection $g\colon V(H) \to \{1, 2, \ldots, n\}$ such that*

$$f^{-1}(t_{v,nm+1}) = (g(v) - 1)\alpha + nm + 1 \text{ for any } v \in V(H). \tag{1}$$

PROOF. Let $f\colon V \to T$ be a block-aligned token placement. Since $f^{-1}(t_{v,1})$, $f^{-1}(t_{v,2}), \ldots, f^{-1}(t_{v,\alpha})$ appear consecutively in G for every $v \in V(H)$, there exists a bijection $g\colon V(H) \to \{1, 2, \ldots, n\}$ such that

$$\{f^{-1}(t_{v,1}), f^{-1}(t_{v,2}), \ldots, f^{-1}(t_{v,\alpha})\} = \{(g(v) - 1)\alpha + 1, (g(v) - 1)\alpha + 2, \ldots, g(v)\alpha\}$$

for $v \in V(H)$. Since $f^{-1}(t_{v,1}), f^{-1}(t_{v,2}), \ldots, f^{-1}(t_{v,\alpha})$ appear in this order or in the reverse order, $f^{-1}(t_{v,nm+1})$ has to be located in the middle of them in either case, where we note that $nm + 1 = (\alpha + 1)/2$. Therefore, $f^{-1}(t_{v,nm+1}) = (g(v) - 1)\alpha + nm + 1$. □

We say that a block-aligned token placement $f\colon V \to T$ *corresponds to* a bijection $g\colon V(H) \to \{1, 2, \ldots, n\}$ if Equation (1) holds.

LEMMA 4. *Let $g_1, g_2\colon V(H) \to \{1, 2, \ldots, n\}$ be bijections with $g_1 \neq g_2$. Suppose that f_1 and f_2 are block-aligned token placements that correspond to g_1 and g_2, respectively. Then, any swap sequence from f_1 and f_2 is of length at least α^2.*

PROOF. Since $g_1 \neq g_2$, there exists a vertex $v \in V(H)$ with $g_1(v) > g_2(v)$. Since

$$\left\{f_1^{-1}(t_{v,1}), f_1^{-1}(t_{v,2}), \ldots, f_1^{-1}(t_{v,\alpha})\right\} = \{(g_1(v) - 1)\alpha + 1, (g_1(v) - 1)\alpha + 2, \ldots, g_1(v)\alpha\}$$

and

$$\left\{f_2^{-1}(t_{v,1}), f_2^{-1}(t_{v,2}), \ldots, f_2^{-1}(t_{v,\alpha})\right\} = \{(g_2(v) - 1)\alpha + 1, (g_2(v) - 1)\alpha + 2, \ldots, g_2(v)\alpha\},$$

the length of any swap sequence from f_1 and f_2 is at least

$$\sum_{i=1}^{\alpha} \left(f_1^{-1}(t_{v,i}) - f_2^{-1}(t_{v,i})\right) = \alpha^2(g_1(v) - g_2(v)) \geq \alpha^2,$$

which completes the proof. □

Suppose that there exists a swap sequence $\mathbf{f}$ of length less than $2\beta(k\alpha + \alpha - m)$ that realizes R^β. Then, there exists a subsequence $\mathbf{f}'$ of $\mathbf{f}$ such that $\mathbf{f}'$ realizes R and $|\mathbf{f}'| \leq |\mathbf{f}|/\beta < 2(k\alpha + \alpha - m)$. Since $|\mathbf{f}'| < 2(k + 1)\alpha \leq \gamma$, if a subsequence of $\mathbf{f}'$ realizes Q^γ, then it contains a token placement that realizes Q. With this observation, we see that $\mathbf{f}'$ contains token placements $f_1^*, f_{e_1}, f_2^*, f_{e_2}, f_3^*, \ldots, f_{e_m}, f_{m+1}^*$ in this order, where f_i^* realizes Q (i.e., it is block-aligned) and f_{e_i} realizes $\psi(e_i)$ for each i.

For $i \in \{1, 2, \ldots, m + 1\}$, Lemma 3 shows that f_i^* corresponds to some bijection $g_i\colon V(H) \to \{1, 2, \ldots, n\}$. Furthermore, since $|\mathbf{f}'| < 2(k + 1)\alpha \leq \alpha^2$, Lemma 4 shows that $g_1 = g_2 = \cdots = g_{m+1}$. That is, $f_1^*, f_2^*, \ldots, f_{m+1}^*$ correspond to a common bijection $g\colon V(H) \to \{1, 2, \ldots, |V(H)|\}$.

We now show that g is a desired bijection. For every $e_i = \{u, v\}$, any swap sequence from f_i^* to f_{e_i} has length at least $|(f_i^*)^{-1}(t_{u,nm+1}) - (f_i^*)^{-1}(t_{v,nm+1})| - 1 = \alpha|g(u) - g(v)| - 1$ as f_i^* corresponds to g. Similarly, any swap sequence from f_{e_i} to f_{i+1}^* has length at least $|(f_{i+1}^*)^{-1}(t_{u,nm+1}) - (f_{i+1}^*)^{-1}(t_{v,nm+1})| - 1 = \alpha|g(u) - g(v)| - 1$. Therefore, we obtain

$$|\mathbf{f}'| \geq \sum_{\{u,v\} \in E(H)} 2(\alpha|g(u) - g(v)| - 1).$$

This together with $|\mathbf{f}'| < 2(k\alpha + \alpha - m)$ shows that $\sum_{\{u,v\}\in E(H)} |g(u) - g(v)| < k + 1$. This implies that $\sum_{\{u,v\}\in E(H)} |g(u) - g(v)| \leq k$ by integrality, and hence g is a desired bijection. This completes the proof of Theorem 1. □

3 Algorithm Parameterized by the Number of Gates

In this section, we assume that $G = (V, E)$ is a path. Let $k = |S|$ be the size of a poset $P = (S, \preceq)$. The purpose of this section is to design a fixed-parameter algorithm for the problem parameterized by k. Since G is a path, we suppose for simplicity that $V = \{1, 2, \ldots, n\}$ and $E = \{\{i, i+1\} \mid i = 1, 2, \ldots, n-1\}$.

We first observe that we may assume that a poset forms a chain. Indeed, suppose that P is not a chain. Then, if we have a fixed-parameter algorithm for a chain, we can apply the algorithm for all the linear extensions of P. Since the number of the linear extensions is at most $k!$, it is a fixed-parameter algorithm for P. Thus, we may assume that $S = \{s_1, s_2, \ldots, s_k\}$ such that $s_1 \prec s_2 \prec \cdots \prec s_k$. Let $\tilde{T} = \bigcup_{s\in S} \varphi(s)$ and let $\tilde{k} = |\tilde{T}|$. Then, we have $\tilde{k} \leq 2k$. We denote $[\tilde{k}] = \{1, 2, \ldots, \tilde{k}\}$.

Let f be a token placement. Let $\tilde{V}$ be the positions where tokens in $\tilde{T}$ are placed, i.e., $\tilde{V} = \{f^{-1}(t) \mid t \in \tilde{T}\}$. We denote $\tilde{V} = \{v_1, v_2, \ldots, v_{\tilde{k}}\}$ where $v_1 < v_2 < \cdots < v_{\tilde{k}}$. Define a vector $x \in \mathbb{Z}^{\tilde{k}}$ so that $x_i = v_{i+1} - v_i$, where $v_0 = 0$, for every index $i = 0, 1, \ldots, \tilde{k}-1$. We note that $\sum_{i=0}^{\tilde{k}-1} x_i = v_{\tilde{k}} \leq n$ holds. We further define a bijection $\sigma\colon [\tilde{k}] \to \tilde{T}$ as $\sigma(i) = f(v_i)$. Let Σ denote the set of all bijections from $[\tilde{k}]$ to $\tilde{T}$. We call the pair (x, σ) the *signature* of the token placement f, which is denoted by $\mathrm{sig}(f)$. The signature maintains the information only on the tokens in $\tilde{T}$, which suffices for finding a shortest feasible swap sequence, since swapping two tokens not in $\tilde{T}$ is redundant in the swap sequence.

We first present a polynomial-time algorithm when k is a fixed constant in Section 3.1, and then a fixed-parameter algorithm in Section 3.2.

3.1 Polynomial-time Algorithm for a Fixed Constant k

Define $\mathcal{R} = \{x \in \mathbb{Z}^{\tilde{k}} \mid \sum x_j \leq n, x_j \geq 1\ (j = 0, 1, \ldots, \tilde{k} - 1)\}$ and $\mathcal{S} = \{(x, \sigma) \mid x \in \mathcal{R}, \sigma \in \Sigma\}$. We see that, for any $(x, \sigma) \in \mathcal{S}$, there exists a token placement f such that $\mathrm{sig}(f) = (x, \sigma)$, and such a placement f can be found in polynomial time even when k is not a constant. It holds that $|\mathcal{S}| = O(\tilde{k}! n^{\tilde{k}})$.

For two signatures (x^0, σ^0) and (x^t, σ^t), a *swap sequence from* (x^0, σ^0) *to* (x^t, σ^t) means a swap sequence $f_0 \rightsquigarrow f_1 \rightsquigarrow \cdots \rightsquigarrow f_\ell$ for some token placements f_0 and f_ℓ such that $\mathrm{sig}(f_0) = (x^0, \sigma^0)$ and $\mathrm{sig}(f_\ell) = (x^t, \sigma^t)$. Its length is defined to be ℓ.

We first show that we can find a shortest swap sequence between two signatures in polynomial time when k is a fixed constant.

Lemma 5. *For two signatures (x^0, σ^0) and (x^t, σ^t), we can find a shortest swap sequence from (x^0, σ^0) to (x^t, σ^t) in $O(\mathrm{poly}(|\mathcal{S}|))$ time.*

Proof. We construct a graph on the vertex set $\mathcal{S}$ such that (x, σ) and (x', σ') are adjacent if and only if there exist two token placements f, f' such that $\mathrm{sig}(f) = (x, \sigma)$, $\mathrm{sig}(f') = (x', \sigma')$, and $f \rightsquigarrow f'$. Then, a path from (x^0, σ^0) to (x^t, σ^t) in the graph corresponds to a swap sequence from (x^0, σ^0) to (x^t, σ^t). Therefore, we can find a shortest swap sequence by finding a shortest path from (x^0, σ^0) to (x^t, σ^t) in the graph. Since the number of vertices of the constructed graph is $|\mathcal{S}|$, it can be done in $O(\mathrm{poly}(|\mathcal{S}|))$ time. □

The above lemma allows us to design a polynomial-time algorithm for a fixed constant k.

Theorem 2. *Let $P = (S, \preceq)$ be a chain of k elements, where k is a fixed constant. For a token placement f_0, we can find a shortest feasible swap sequence from f_0 in polynomial time.*

Proof. Let $\mathcal{S}_i = \{(x,\sigma) \in \mathcal{S} \mid \exists f \text{ s.t. } \mathrm{sig}(f) = (x,\sigma) \text{ and } f^{-1}(\varphi(s_i)) \in E\}$, which is the set of signatures (x,σ) that correspond to token placements in which $\varphi(s_i)$ are adjacent. Note that if $(x,\sigma) \in \mathcal{S}_i$, then $\mathrm{sig}(f) = (x,\sigma)$ implies $f^{-1}(\varphi(s_i)) \in E$ for any f, because $\varphi(s_i) \subseteq \tilde{T}$. Moreover, let $\mathcal{S}_0 = \{\mathrm{sig}(f_0)\}$.

Define a digraph $\mathcal{G} = (\bigcup_{i=0}^{k} \mathcal{S}_i, \bigcup_{i=0}^{k-1} E_i)$, where $E_i = \{((x,\sigma),(x',\sigma')) \mid (x,\sigma) \in \mathcal{S}_i, (x',\sigma') \in \mathcal{S}_{i+1}\}$ for $i = 0,1,\dots,k-1$. We suppose that $e = ((x,\sigma),(x',\sigma'))$ in E_i has a length equal to the shortest length of a swap sequence from (x,σ) to (x',σ'), which can be computed in polynomial time by Lemma 5.

We see that the shortest path from the vertex in $\mathcal{S}_0$ to some vertex in $\mathcal{S}_k$ corresponds to a shortest feasible swap sequence from f_0. The number of vertices of the graph is bounded by $O(k|\mathcal{S}|)$, which is polynomial when k is a constant. Thus, the theorem holds. □

3.2 Fixed-parameter Algorithm

In this section, we present a fixed-parameter algorithm parameterized by k by dynamic programming.

We first observe that we can compute the shortest length of a swap sequence between two signatures with the same bijection σ.

Lemma 6. *Suppose that we are given two signatures (x,σ) and (y,σ) with the same bijection σ. Then, the shortest length of a swap sequence from (x,σ) to (y,σ) is equal to $\sum_{i=1}^{\tilde{k}} |v_i - w_i|$, where $v_i = \sum_{j=0}^{i-1} x_j$ and $w_i = \sum_{j=0}^{i-1} y_j$ for $i = 1,2,\dots,\tilde{k}$. Moreover, there exists a shortest swap sequence such that all the token placements in the sequence have the same bijection σ in their signatures.*

Proof. Since the initial and target token placements have the same σ in their signature, we need $|w_i - v_i|$ swaps to move the token $\sigma(i)$ to w_i for any $i = 1,2,\dots,\tilde{k}$. Thus, the shortest swap sequence has length at least $\sum_{i=1}^{\tilde{k}} |v_i - w_i|$. To see that they are equal, we show the existence of a swap sequence of length $\sum_{i=1}^{\tilde{k}} |v_i - w_i|$ by induction on this value. If $\sum_{i=1}^{\tilde{k}} |v_i - w_i| = 0$, then the claim is obvious. Otherwise, let p be the minimum index such that $v_p \neq w_p$. By changing the roles of x and y if necessary, we may assume that $v_p > w_p$. Then, starting from (x,σ), we apply swap operations $v_p - w_p$ times to obtain a new signature (x',σ) such that $x'_{p-1} = x_{p-1} - (v_p - w_p)$ and $x'_p = x_p + (v_p - w_p)$. That is, $v'_i = v_i$ for $i \in [\tilde{k}] \setminus \{p\}$ and $v'_p = w_p$, where v'_i is defined as $v'_i = \sum_{j=0}^{i-1} x'_j$. Note that this operation is possible without changing the bijection σ, because $v'_p = w_p > w_{p-1} = v_{p-1}$ by the minimality of p. By the induction hypothesis, there exists a swap sequence of length $\sum_{i=1}^{\tilde{k}} |v'_i - w_i|$ between (x',σ) and (y,σ). Therefore, we obtain a swap sequence between (x,σ) and (y,σ) whose length is $|v_p - w_p| + \sum_{i=1}^{\tilde{k}} |v'_i - w_i| = \sum_{i=1}^{\tilde{k}} |v_i - w_i|$. Moreover, each token placement in the obtained swap sequence has the same bijection σ. □

By the lemma, the shortest length of a swap sequence from (x,σ) to (y,σ) does not depend on σ. Thus, we denote it by $d(x,y)$ for $x,y \in \mathcal{R}$.

Let $\mathbf{f}$ be a feasible swap sequence $f_0 \rightsquigarrow f_1 \rightsquigarrow \cdots \rightsquigarrow f_\ell$. Suppose that $\varphi(s_j)$ is realized at token placement f_{i_j}, that is, $f_{i_j}^{-1}(\varphi(s_j)) \in E$ for $j = 1,2,\dots,k$ and $i_1 \leq i_2 \leq \cdots \leq i_k$. We define $i_0 = 0$. Note that the number of distinct values in $\{i_1,i_2,\dots,i_k\}$, denoted by α, is at most k. Also, let β be the number of times that σ in the signature changes in the swap sequence. We call $\alpha + \beta$ the *signature length* of $\mathbf{f}$.

The following lemma says that the signature length is bounded by a function of k, which we denote by $\ell_{\max}$.

Lemma 7. *For a shortest feasible swap sequence $f_0 \rightsquigarrow f_1 \rightsquigarrow \cdots \rightsquigarrow f_\ell$, the signature length is bounded by $((2k)! + 1)k$ from above.*

Proof. Consider the partial swap sequence $f_{i_{j-1}} \rightsquigarrow \cdots \rightsquigarrow f_{i_j}$ for $j = 1, 2, \ldots, k$. We observe that, in the partial swap sequence, the number of times that bijections change is at most $\tilde{k}!$. Indeed, in this partial sequence, token placements with the same bijection appear sequentially, as otherwise, we can short-cut between them by Lemma 6. Therefore, the total signature length is bounded by $k(\tilde{k}!) + k \leq k((2k)!) + k$. □

Let $g^\ell(x, \sigma, P)$ be the shortest length of a swap sequence to realize P from some token placement f_0 with $\mathrm{sig}(f_0) = (x, \sigma)$ such that it has signature length at most ℓ. In what follows, we derive a recursive equation on $g^\ell(x, \sigma, P)$ for dynamic programming.

We first give notation. For a bijection $\sigma \in \Sigma$ and a non-negative integer j with $1 \leq j \leq \tilde{k} - 1$, let $\sigma_j \in \Sigma$ be the bijection obtained from σ by swapping the jth and $(j+1)$-st tokens. We define $\mathcal{R}_j = \{x \in \mathcal{R} \mid x_j = 1\}$, which is the set of signatures such that the jth token and the $(j+1)$-st token are adjacent.

To derive a recursive equation on $g^\ell(x, \sigma, P)$, consider the following two cases in which the signature length is decreased at least by one, separately.

The first case is when the bijection σ is changed, i.e., β decreases. Suppose that we change σ to σ_j for some $1 \leq j \leq \tilde{k} - 1$. In this case, we first move to (x', σ) for some $x' \in \mathcal{R}_j$, and then change (x', σ) to (x', σ_j). By Lemma 6, the number of swaps is $d(x, x') + 1$. After moving to (x', σ_j), we can recursively consider finding a shortest swap sequence to realize P from (x', σ_j) with the signature length at most $\ell - 1$. Therefore, the total length in this case is $g^{\ell-1}(x', \sigma_j, P) + d(x, x') + 1$, and hence the shortest length when we change σ is equal to

$$\min_{1 \leq j \leq \tilde{k}-1} \min_{x' \in \mathcal{R}_j} \left\{ g^{\ell-1}(x', \sigma_j, P) + d(x, x') + 1 \right\}.$$

The other case is when s_1 is realized without changing σ, i.e., α decreases. Then, it is necessary that $s_1 = (\sigma(h), \sigma(h+1))$ for some $1 \leq h \leq \tilde{k} - 1$. To realize s_1, we move (x, σ) to (x', σ) for some $x' \in \mathcal{R}_h$. By recursion, the total length in this case is $g^{\ell-1}(x', \sigma, P') + d(x, x')$ by Lemma 6, where P' is the poset obtained from P by removing the first element s_1, that is, P' forms the chain $s_2 \prec s_3 \prec \cdots \prec s_k$. Thus, the shortest length in this case is

$$\min_{x' \in \mathcal{R}_h} \left\{ g^{\ell-1}(x', \sigma, P') + d(x, x') \right\}.$$

In summary, we have that, for any $x \in \mathcal{R}$, $\sigma \in \Sigma$, and $1 \leq \ell \leq \ell_{\max}$,

$$g^\ell(x, \sigma, P) = \min \Bigg\{ \min_{1 \leq j \leq \tilde{k}-1} \min_{x' \in \mathcal{R}_j} \left\{ g^{\ell-1}(x', \sigma_j, P) + d(x, x') + 1 \right\}, \min_{x' \in \mathcal{R}_h} \left\{ g^{\ell-1}(x', \sigma, P') + d(x, x') \right\} \Bigg\}, \tag{2}$$

where $s_1 = (\sigma(h), \sigma(h+1))$ for some h. If such h does not exist, then the second term is defined to be $+\infty$.

It follows from Equation (2) that we can design a dynamic programming algorithm. However, the running time would become $O(k \cdot k! \ell_{\max} |\mathcal{R}|)$, and this does not give a fixed-parameter algorithm, since $|\mathcal{R}| = O(n^{\tilde{k}})$. In what follows, we will reduce the running time by showing that the minimum is achieved at an extreme point.

For $i = 0, 1, \ldots \tilde{k} - 1$, let e_i denote the unit vector whose ith entry is one and the other entries are zeros. For a vector $x \in \mathcal{R}$ and $1 \leq j \leq \tilde{k} - 1$, define $N_j(x)$ as

$$N_j(x) = \{x + a e_{j-1} - (x_j - 1) e_j + b e_{j+1} \mid a + b = x_j - 1,\ a, b \in \mathbb{Z}_+\},$$

where we regard $e_{\tilde{k}}$ as the zero vector to simplify the notation. Then, $x' \in N_j(x)$ satisfies that

$$\begin{aligned} x'_j &= 1, \\ x'_{j-1} + x'_{j+1} &= x_{j-1} + x_j + x_{j+1} - 1, \\ x'_i &= x_i \text{ for } i \notin \{j-1, j, j+1\}, \end{aligned}$$

where $x_{\tilde{k}} = n - \sum_{i=0}^{\tilde{k}-1} x_i$ and $x'_{\tilde{k}} = n - \sum_{i=0}^{\tilde{k}-1} x'_i$. The signature (x', σ) with $x' \in N_j(x)$ means that it is obtained from (x, σ) by only moving two tokens $\sigma(j)$ and $\sigma(j+1)$ so that the two tokens are adjacent. Moreover, define y^j and y'^j to be vectors in $N_j(x)$ in which $(a, b) = (0, x_j - 1)$ and $(a, b) = (x_j - 1, 0)$, respectively. Then,

$$\begin{aligned} (y^j)_{j-1} &= x_{j-1}, \quad (y^j)_{j+1} = x_j + x_{j+1} - 1, \\ (y'^j)_{j-1} &= x_{j-1} + x_j - 1, \quad (y'^j)_{j+1} = x_{j+1}. \end{aligned}$$

Thus, (y^j, σ) ((y'^j, σ), respectively) is obtained from (x, σ) by only moving one token $\sigma(j+1)$ ($\sigma(j)$, respectively) so that $\sigma(j)$ and $\sigma(j+1)$ are adjacent.

The following theorem asserts that the minimum of Equation (2) is achieved at either y^j or y'^j.

Theorem 3. *For any $x \in \mathcal{R}$, $\sigma \in \Sigma$, and $1 \le \ell \le \ell_{\max}$, it holds that*

$$g^\ell(x, \sigma, P) = \min\Bigg\{ \min_{1 \le j \le \tilde{k}-1,\, y \in \{y^j, y'^j\}} \left\{ g^{\ell-1}(y, \sigma_j, P) + d(x, y) + 1 \right\}, \\ \min_{y \in \{y^h, y'^h\}} \left\{ g^{\ell-1}(y, \sigma, P') + d(x, y) \right\} \Bigg\}.$$

To prove the theorem, we show the following two lemmas. We first show that the minimum of Equation (2) is achieved at some point in $N_j(x)$.

Lemma 8. *It holds that, for any $x \in \mathcal{R}$, $\sigma \in \Sigma$, $1 \le \ell \le \ell_{\max}$ and $1 \le j \le \tilde{k} - 1$,*

$$\min_{x' \in \mathcal{R}_j} \left\{ g^{\ell-1}(x', \sigma, P) + d(x, x') \right\} = \min_{y \in N_j(x)} \left\{ g^{\ell-1}(y, \sigma, P) + d(x, y) \right\}.$$

Proof. Let x^* be a vector in $\mathcal{R}_j$ that attains the minimum of the LHS, that is,

$$g^{\ell-1}(x^*, \sigma, P) + d(x, x^*) = \min_{x' \in \mathcal{R}_j} \left\{ g^{\ell-1}(x', \sigma, P) + d(x, x') \right\}.$$

To simplify the notation, we denote $g^* = g^{\ell-1}(x^*, \sigma, P)$ and $g_y = g^{\ell-1}(y, \sigma, P)$ for a vector $y \in N_j(x)$.

Since $\mathcal{R}_j \supseteq N_j(x)$, it holds that, for any $y \in N_j(x)$,

$$g^* + d(x, x^*) \le g_y + d(x, y). \tag{3}$$

The minimality of g_y implies that $g_y \le g^* + d(y, x^*)$. Hence, it holds by Equation (3) that, for any $y \in N_j(x)$,

$$g^* + d(x, x^*) \le g_y + d(x, y) \le g^* + d(x, y) + d(y, x^*). \tag{4}$$

To show the lemma, it suffices to find $y \in N_j(x)$ that satisfies Equation (4) with equality.

By definition, $d(x, y) = x_j - 1$ for any $y \in N_j(x)$. We denote $v_i = \sum_{h=0}^{i-1} x_h$, $v_i^* = \sum_{h=0}^{i-1} x_h^*$, and $w_i = \sum_{h=0}^{i-1} y_h$ for any $i = 1, 2, \ldots, \tilde{k}$. Then, Lemma 6 implies that

$$d(x, x^*) = \sum_{i=1}^{\tilde{k}} |v_i - v_i^*| = X + |v_j - v_j^*| + |v_{j+1} - v_{j+1}^*| + X',$$

where we define $X = \sum_{i=1}^{j-1} |v_i - v_i^*|$ and $X' = \sum_{i=j+2}^{\tilde{k}} |v_i - v_i^*|$. We distinguish three cases by the position of v_j^*. Note that $v_{j+1}^* = v_j^* + 1$, since $x^* \in \mathcal{R}_j$.

First, suppose that v_j^* satisfies that $v_j < v_j^* < v_{j+1}$. Then, since $|v_j - v_j^*| + |v_{j+1} - v_{j+1}^*| = v_j^* - v_j + v_{j+1} - (v_j^* + 1) = x_j - 1$, it holds that $d(x, x^*) = X + x_j - 1 + X'$. We define $y \in N_j(x)$ so that $w_j = v_j^*$ and $w_{j+1} = v_{j+1}^*$. Then, we have $d(y, x^*) = X + X'$, since $w_i = v_i$ for any $i \in \{1, 2, \ldots, j-1\} \cup \{j+2, j+3, \ldots, \tilde{k}\}$. Therefore, since $d(x, y) = x_j - 1$, we obtain $d(x, x^*) = d(y, x^*) + d(x, y)$. Thus, Equation (4) holds with equality.

Next, suppose that v_j^* satisfies that $v_j^* \leq v_j$. Then, it holds that

$$\begin{aligned} d(x, x^*) &= X + X' + |v_j - v_j^*| + |v_{j+1} - v_{j+1}^*| \\ &= X + X' + (v_j - v_j^*) + (v_{j+1} - v_j + v_j - (v_j^* + 1)) \\ &= X + X' + 2(v_j - v_j^*) + x_j - 1, \end{aligned}$$

since $v_{j+1} - v_j = x_j$. We define $y \in N_j(x)$ so that $w_j = v_j$ and $w_{j+1} = v_j + 1$. Then, we have $d(y, x^*) = X + X' + 2(v_j - v_j^*)$, since $w_i = v_i$ for any $i \in \{1, 2, \ldots, j-1\} \cup \{j+2, j+3, \ldots, \tilde{k}\}$. Therefore, since $d(x, y) = x_j - 1$, we obtain $d(x, x^*) = d(y, x^*) + d(x, y)$, and hence Equation (4) holds with equality.

The case where $v_j^* \geq v_{j+1}$ is analogous to the second case. Therefore, in any case, there exists $y \in N_j(x)$ that satisfies Equation (4) with equality. Thus, the lemma holds. □

By Lemma 8, it holds that

$$g^\ell(x, \sigma, P) = \min \left\{ \min_{1 \leq j \leq \tilde{k}-1} \min_{y \in N_j(x)} \left\{ g^{\ell-1}(y, \sigma_j, P) + x_j \right\}, \min_{y \in N_h(x)} \left\{ g^{\ell-1}(y, \sigma, P') + x_h - 1 \right\} \right\}, \quad (5)$$

in which we note $d(x, y) = x_j - 1$ for $y \in N_j(x)$.

Lemma 9. *The function $g^\ell(x, \sigma, P)$ can be expressed as the minimum of linear functions on x. That is, for any $\sigma \in \Sigma$ and $1 \leq \ell \leq \ell_{\max}$, there exist vectors $c_1, c_2, \ldots, c_p$ and real numbers $\delta_1, \delta_2, \ldots, \delta_p$ such that*

$$g^\ell(x, \sigma, P) = \min_{1 \leq i \leq p} \{c_i x + \delta_i\}$$

for any $x \in \mathcal{R}$.

Proof. We prove this lemma by induction on ℓ. In the base case where $\ell = 0$, the poset P is empty, and σ is not changed. Thus, $g^0(x, \sigma, P) = 0$. Suppose that $\ell \geq 1$.

By Equation (5), it suffices to show that, for any j and σ',

$$\min_{y \in N_j(x)} \left\{ g^{\ell-1}(y, \sigma', P) + x_j \right\}$$

can be expressed as the minimum of linear functions. By the induction hypothesis, $g^{\ell-1}$ is the minimum of linear functions. Hence, the above can be expressed as

$$\min_{y \in N_j(x)} \left\{ \min_{1 \leq i \leq p'} \{c_i' y + \delta_i'\} + x_j \right\} = \min_{1 \leq i \leq p'} \min_{y \in N_j(x)} \left\{ c_i' y + x_j + \delta_i' \right\}$$

for some linear functions $c_i' y + \delta_i'$ for $i = 1, 2, \ldots, p'$.

Recall that, for every $y \in N_j(x)$, we can write $y = x + a e_{j-1} - (x_j - 1) e_j + b e_{j+1}$ for some $a, b \in \mathbb{Z}_+$ with $a + b = x_j - 1$. Hence, it holds that

$$c_i' y + x_j + \delta_i' = c_i'(x + a e_{j-1} - (x_j - 1) e_j + b e_{j+1}) + x_j + \delta_i',$$

which is a linear function on x for given a, b, and j. Therefore, the lemma holds by induction. □

With Lemma 9, we are ready to prove Theorem 3.

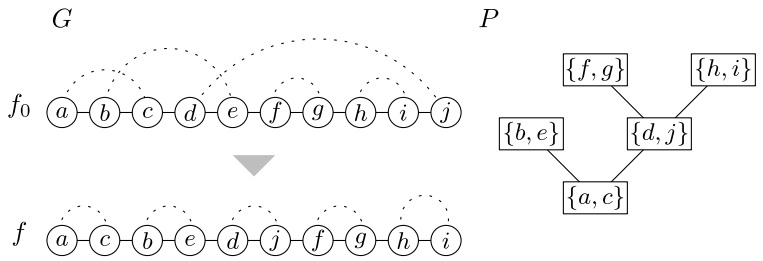


Fig. 3. The left figure is an initial assignment f_0 of tokens on a path G, and the right figure is a poset P.

Proof of Theorem 3. By Equation (5), it suffices to show that, for any $\sigma \in \Sigma$ and $1 \le j \le \tilde{k}-1$, either y^j or y'^j achieves the minimum of

$$\min_{y \in N_j(x)} \left\{ g^{\ell-1}(y, \sigma, P) + x_j \right\}.$$

Since $g^{\ell-1}$ is in the form of the minimum of linear functions by Lemma 9, the above can be expressed as

$$\min_{1 \le i \le p'} \min_{a,b \in \mathbb{Z}_+ : a+b = x_j - 1} \left\{ c_i'(x + ae_{j-1} - (x_j - 1)e_j + be_{j+1}) + x_j + \delta_i' \right\}$$

for some linear functions $c_i' y + \delta_i'$ for $i = 1, 2, \ldots, p'$. For given x and j, the minimum is attained either at $(a, b) = (x_j - 1, 0)$ or at $(a, b) = (0, x_j - 1)$. They correspond to y^j and y'^j, respectively. Thus, the theorem holds. □

Let f_0 be the initial placement and $\mathrm{sig}(f_0) = (x^0, \sigma^0)$. Let $v_i^0 = \sum_{j=0}^{i-1} x_j^0$ for each i. For $i = 0, 1, \ldots, \tilde{k}-1$, define $B_i = \{f_0(v_i^0 + 1), f_0(v_i^0 + 2), \ldots, f_0(v_{i+1}^0 - 1)\}$, which are tokens not in $\tilde{T}$. Then, $|B_i| = x_i^0 - 1$. Theorem 3 shows that, when we compute $g^\ell(x, \sigma, P)$ by using the formula, it suffices to consider signatures (y, σ') such that all tokens in B_i appear consecutively along the path. That is, we only consider vectors y such that, for each i, $y_i - 1 = \sum_{h=i'}^{j'} |B_h|$ for some $0 \le i', j' \le \tilde{k}-1$ (possibly, $y_i - 1 = 0$). This shows that each y_i can take one of $O(\tilde{k}^2)$ values, and hence y has $O(\tilde{k}^{2\tilde{k}})$ choices. Therefore, the size of the DP table is $O(k \cdot k! \ell_{\max} \tilde{k}^{2\tilde{k}})$, and each step can be processed in a fixed-parameter time. Thus, we have the following theorem.

Theorem 4. *There exists a fixed-parameter algorithm for Qubit Routing when G is a path and $k = |S|$ is a parameter.* □

4 Polynomial-time Algorithm for Disjoint Two-qubit Operations

We say that an instance $(G, P = (S, \preceq), T, \varphi, f_0)$ of Qubit Routing has *disjoint pairs* if $\varphi(s) \cap \varphi(s') = \emptyset$ for any pair of distinct elements $s, s' \in S$. See Figure 3 for an example. The objective of this section is to give a polynomial-time algorithm for instances with disjoint pairs when the graph G is a path.

Theorem 5. *Qubit Routing can be solved in polynomial time when a given graph is a path and the instance has disjoint pairs.*

Let $(G, P = (S, \preceq), T, \varphi, f_0)$ be an instance with disjoint pairs. Since G is a path, we suppose for simplicity that $V = \{1, 2, \ldots, n\}$ and $E = \{\{i, i+1\} \mid i = 1, 2, \ldots, n-1\}$. Let f be a token placement. For an element $s \in S$ with $f^{-1}(\varphi(s)) = \{\alpha_1, \alpha_2\}$, $\alpha_1 < \alpha_2$, we define $\mathrm{gap}_f(s) := \alpha_2 - \alpha_1 - 1$. Then, $\mathrm{gap}_f(s)$ is a lower bound on the number of swaps to realize $\varphi(s)$ if the initial token placement is f. For distinct elements $s, s' \in S$ such that $f^{-1}(\varphi(s)) = \{\alpha_1, \alpha_2\}$, $\alpha_1 < \alpha_2$, and $f^{-1}(\varphi(s')) = \{\beta_1, \beta_2\}$, $\beta_1 < \beta_2$, we say that s and s' *cross* if $\alpha_1 < \beta_1 < \alpha_2 < \beta_2$ or $\beta_1 < \alpha_1 < \beta_2 < \alpha_2$. One can

observe that, if $S = \{s, s'\}$ such that s and s' cross, then we can realize both $\varphi(s)$ and $\varphi(s')$ by $\mathrm{gap}_f(s) + \mathrm{gap}_f(s') - 1$ swaps. Note that $\alpha_1, \alpha_2, \beta_1$, and β_2 are always distinct, since the instance has disjoint pairs. We define

$$\begin{aligned}\mathrm{gap}(f) &:= \sum_{s \in S} \mathrm{gap}_f(s),\\ \mathrm{cross}(f) &:= \left|\left\{\{s, s'\} \in \binom{S}{2} \;\middle|\; s \text{ and } s' \text{ cross}\right\}\right|,\\ \mathrm{value}(f) &:= \mathrm{gap}(f) - \mathrm{cross}(f),\end{aligned}$$

where gap, cross, and value are regarded as functions only in f by fixing G, P, T, and φ. For example, we see in Figure 3 that $\mathrm{gap}_{f_0}(\{a, c\}) = 1$ and $\mathrm{gap}_{f_0}(\{b, e\}) = 2$. Moreover, it holds that $\mathrm{gap}(f_0) = 1+2+5+0+0 = 8$. Also, since $\mathrm{cross}(f_0) = 2$, we have $\mathrm{value}(f_0) = 8-2 = 6$ by definition. However, it holds that $\mathrm{value}(f) = \mathrm{gap}(f) = \mathrm{cross}(f) = 0$.

We prove the following proposition, which completes the proof of Theorem 5.

Proposition 2. *The optimal value for the instance* $(G, P = (S, \preceq), T, \varphi, f_0)$ *is* $\mathrm{value}(f_0)$. *Such a sequence can be constructed in polynomial time.*

To prove Proposition 2, we first show that the optimal value is at most $\mathrm{value}(f_0)$.

Lemma 10. *The instance* $(G, P = (S, \preceq), T, \varphi, f_0)$ *has a feasible swap sequence of length* $\mathrm{value}(f_0)$, *and such a swap sequence can be found in polynomial time.*

Proof. We prove the statement by induction on the lexicographic ordering of $(\mathrm{gap}(f_0), \mathrm{value}(f_0))$. Note that $\mathrm{gap}(f_0) \geq 0$ and $\mathrm{value}(f_0) \geq -|\binom{S}{2}|$.[1]

If $\mathrm{gap}(f_0) = 0$, then $f_0^{-1}(\varphi(x))$ forms a pair of adjacent vertices for any $x \in S$, and hence $\mathrm{cross}(f_0) = \mathrm{value}(f_0) = 0$ holds. In such a case, since the trivial swap sequence consisting of a single token placement f_0 is a feasible sequence of length zero, the statement holds.

Suppose that $\mathrm{gap}(f_0) \geq 1$. Let

$$i^* := \max\{i \in \{1, 2, \ldots, n\} \mid \exists s^* \in S \text{ s.t. } f_0^{-1}(\varphi(s^*)) = \{i, j\} \text{ with } j \geq i + 2\},$$

which is well-defined as $\mathrm{gap}(f_0) \geq 1$. Let f_1 be the token placement obtained from f_0 by applying a swap operation on $\{i^*, i^*+1\} \in E$. Then, we see that $\mathrm{value}(f_1) = \mathrm{value}(f_0) - 1$ and $\mathrm{gap}(f_1) \leq \mathrm{gap}(f_0)$ by the following case analysis.

(1) Suppose that $i^* + 1 \notin f_0^{-1}(\varphi(s))$ for any $s \in S$. In this case, $\mathrm{gap}(f_1) = \mathrm{gap}(f_0) - 1$ and $\mathrm{cross}(f_1) = \mathrm{cross}(f_0)$, which shows that $\mathrm{value}(f_1) = \mathrm{value}(f_0) - 1$.
(2) Suppose that $i^*+1 \in f_0^{-1}(\varphi(s))$ for some $s \in S$. By the maximality of i^* and by the assumption that the instance has disjoint pairs, we obtain $f_0^{-1}(\varphi(s)) = \{j, i^* + 1\}$ for $j < i^*$ or $j = i^* + 2$.
   — If $f_0^{-1}(\varphi(s)) = \{j, i^* + 1\}$ for some $j < i^*$, then $\mathrm{gap}(f_1) = \mathrm{gap}(f_0) - 2$ and $\mathrm{cross}(f_1) = \mathrm{cross}(f_0) - 1$, which shows that $\mathrm{value}(f_1) = \mathrm{value}(f_0) - 1$.
   — If $f_0^{-1}(\varphi(s)) = \{i^* + 1, i^* + 2\}$, then $\mathrm{gap}(f_1) = \mathrm{gap}(f_0)$ and $\mathrm{cross}(f_1) = \mathrm{cross}(f_0) + 1$, which shows that $\mathrm{value}(f_1) = \mathrm{value}(f_0) - 1$.

Since $(\mathrm{gap}(f_1), \mathrm{value}(f_1))$ is lexicographically smaller than $(\mathrm{gap}(f_0), \mathrm{value}(f_0))$, by the induction hypothesis, instance $(G, P = (S, \preceq), T, \varphi, f_1)$ has a feasible swap sequence of length $\mathrm{value}(f_1) = \mathrm{value}(f_0) - 1$. Since f_1 is obtained from f_0 by a single swap operation, this shows that $(G, P = (S, \preceq), T, \varphi, f_0)$ has a feasible swap sequence of length $\mathrm{value}(f_0)$. □

[1] Although we can show a better lower bound on $\mathrm{value}(f_0)$ by careful analysis, we just present a trivial lower bound here, because we only need the fact that $\mathrm{value}(f_0)$ has a finite lower bound to apply the induction.

The proof of Lemma 10 shows that, by applying value(f_0) swap operations, we can transform f_0 into another token placement f such that $f^{-1}(\varphi(s))$ forms a pair of adjacent vertices for any $s \in S$. This means that the obtained sequence is a feasible solution for any poset P on S. See the lower-left figure in Figure 3, which is the token placement obtained by the procedure in the proof of Lemma 10.

We next show that the optimal value is at least value(f_0).

Lemma 11. *Each feasible swap sequence for the instance* $(G, P = (S, \preceq), T, \varphi, f_0)$ *has length at least* value(f_0).

Proof. It suffices to show that no single swap operation can decrease value(f) by more than one. Consider a single swap operation on $\{i, i+1\} \in E$ that transforms f into f'. It is easy to see that $\mathrm{gap}(f') \geq \mathrm{gap}(f) - 2$ and $\mathrm{cross}(f') \leq \mathrm{cross}(f) + 1$. We now prove $\mathrm{value}(f') \geq \mathrm{value}(f) - 1$ by the case analysis.

(1) Suppose that $\mathrm{gap}(f') \geq \mathrm{gap}(f)$. In this case, since $\mathrm{cross}(f') \leq \mathrm{cross}(f) + 1$, we obtain $\mathrm{value}(f') \geq \mathrm{value}(f) - 1$.
(2) Suppose that $\mathrm{gap}(f') = \mathrm{gap}(f) - 1$. In this case, we have one of the following.
   — There exists $s \in S$ such that $f^{-1}(\varphi(s)) = \{j, i+1\}$ for some $j \leq i-1$, and $i \notin f^{-1}(\varphi(s'))$ for any $s' \in S$, or
   — There exists $s \in S$ such that $f^{-1}(\varphi(s)) = \{i, j\}$ for some $j \geq i+2$, and $i+1 \notin f^{-1}(\varphi(s'))$ for any $s' \in S$.

   In both cases, we obtain $\mathrm{cross}(f') = \mathrm{cross}(f)$, which shows that $\mathrm{value}(f') = \mathrm{value}(f) - 1$.
(3) Suppose that $\mathrm{gap}(f') = \mathrm{gap}(f) - 2$. In this case, there exist $s, s' \in S$ such that $f^{-1}(\varphi(s)) = \{j, i+1\}$ for some $j \leq i-1$ and $f^{-1}(\varphi(s')) = \{i, k\}$ for some $k \geq i+2$. Then, we obtain $\mathrm{cross}(f') = \mathrm{cross}(f) - 1$, which shows that $\mathrm{value}(f') = \mathrm{value}(f) - 1$.

This completes the proof of the lemma. □

5 Hardness: Stars and Antichains

In this section, we show that the problem is NP-hard even when G is a star and P is an antichain. Recall that an antichain is a poset in which every pair of elements is incomparable.

Theorem 6. *Qubit Routing is NP-hard even when G is a star and P is an antichain.*

Proof. We reduce Vertex Cover, which is known to be NP-hard [5].

> Vertex Cover
> **Input.** A graph $H = (V(H), E(H))$ and a positive integer k.
> **Question.** Is there a vertex subset $X \subseteq V(H)$, called a *vertex cover* of H, such that $|X| \leq k$ and $\{u, v\} \cap X \neq \emptyset$ for every edge $\{u, v\} \in E(H)$?

Suppose that we are given an instance of Vertex Cover that consists of a graph $H = (V(H), E(H))$ and a positive integer k. Let $n = |V(H)|$. We construct an instance of Qubit Routing as follows. Define a set of tokens as $T = V(H) \cup \{0\}$, where $0 \notin V(H)$. Let $S = E(H)$ and define $\varphi\colon S \to \binom{T}{2}$ as $\varphi(\{u, v\}) = \{u, v\}$ for each $\{u, v\} \in S$. The poset P is an antichain, i.e., every pair of elements in S is incomparable. Define a graph $G = (V, E)$ as $V = \{0, 1, \ldots, n\}$ and $E = \{\{0, i\} \mid i \in \{1, 2, \ldots, n\}\}$. The initial token placement $f_0\colon V \to T$ should satisfy $f_0(0) = 0$ and other tokens can be placed arbitrarily over the vertices $\{1, 2, \ldots, n\}$.

We claim that there exists a swap sequence of length k for the instance (G, P, T, φ, f_0) of Qubit Routing if and only if there exists a vertex cover X in H of size k. Suppose that there exists a vertex cover $X = \{v_1, v_2, \ldots, v_k\}$ in H of size k. Then, we construct the following swap sequence.

In the swap $f_{i-1} \rightsquigarrow f_i$, we exchange the token at the vertex 0 and the token v_i. Then, for each edge $\{v_i, v\} \in E(H)$, it holds that $\{f_i^{-1}(v_i), f_i^{-1}(v)\} = \{0, f_i^{-1}(v)\} \in E$. Therefore, the swap sequence $f_0 \rightsquigarrow \cdots \rightsquigarrow f_k$ realizes P. Conversely, suppose that there exists a swap sequence $f_0 \rightsquigarrow f_1 \rightsquigarrow \cdots \rightsquigarrow f_k$ of length k for the instance (G, P, T, φ, f_0). Let $X = \{v \in V(H) \mid f_i^{-1}(0) = v \text{ for some } i \in \{1, 2, \ldots, k\}\}$, i.e., X is the set of vertices of H that are placed at 0 in the course of swaps. Since the swap sequence realizes P, for every edge $\{u, v\} \in E(H)$, there exists f_i such that $\{f_i^{-1}(u), f_i^{-1}(v)\} \in E$, which implies that $u \in X$ or $v \in X$. Therefore, X is a vertex cover of H. □

6 Concluding Remarks

We initiated algorithmic studies on the quantum routing problem, also known as the swap minimization problem, from the viewpoint of theoretical computer science. The problem is of central importance in compiler design for quantum programs when they are implemented in some of the superconducting quantum computers such as IBM Quantum systems.

Most notably, we proved the quantum routing problem is NP-hard even when the graph topology of a quantum computer is a path, which corresponds to the so-called linear nearest neighbor architecture. In our proof, the initial token placement can be chosen arbitrarily. This implies that the combined optimization of the quantum assignment and the quantum routing is also NP-hard for the same architecture.

We also gave some algorithmic results, but they were restricted to the case of the linear nearest neighbor architectures. Possible future work is to give algorithmic results with theoretical guarantees for other graph topologies.

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